LESSON PLAN

AIRCRAFT ARMAMENT (18)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	05
Total No. of Lesson Plans	12	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Introduction to sonar locator beacon
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Introduction to sonar locator beacon. Basic principle of sonar locator beacon.
5. Specific Objectives	18.1.3.1 Trainee will be able to describe the basic theory of operation of surveillance system of an aircraft.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of IB diagrams and L & Ds; Instructor supervision on theory comprehension
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the aircraft surveillance system?
(b) How does the aircraft communicate its position to ground radar?
(c) What are the two main types of transponders used in aviation?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of Sonar Locator Beacon • Role in Aircraft Surveillance System • Basic Theory of Operation • Signal Transmission and Reception • Integration with Transponder System	L & Ds	PPT, IB	15 min
(b)	• Frequency Bands and Allocation • Pulse Modulation Principles • Echo Detection Mechanism • System Components Overview • Operational Constraints and Limits	L & Ds	PPT, PDF	20 min
(c)	• Comparison with Radar Systems • Advantages of Sonar Locator • Limitations in Aviation Context • Future Trends in Surveillance • Safety Implications of Beacon Failure	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the basic principle of operation of the sonar locator beacon.
(b) Describe how the beacon integrates with the aircraft surveillance system.

12. Home Assignment and Follow Up	Prepare a brief note on the differences between the sonar locator beacon and the I-band transponder discussed in LP 06.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Basic principle of operation of I-band Transponder. Frequency used. Interrogation and response format.	Refer to the master syllabus reference list	LP 06

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min